Marco Schouten

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Education

KTH Royal Institute of Technology — Stockholm, Sweden
Master of Science in Machine Learning, Grade 4.84/5.0 — 08/2020 - 10/2022

Delft University of Technology — Delft, Netherlands
Exchange Master in Computer Science — 08/2021 - 01/2022

University of Genoa — Genoa, Italy
Bachelor of Science in Computer Engineering, Grade 109/110 — 09/2017 - 07/2020

Experience

Aptiv PLC — Wuppertal, Germany
Research Assistant — 03/2023 - Present

- Developed and implemented a scalable method based for automatically generating high-fidelity centerlines in high-definition maps, utilizing raw point-cloud data from on-board sensors.
- Conducted initial tests of the developed method, demonstrating its capability to autonomously generate accurate centerlines without human intervention and validating the potential impact of this technology.
- Trained and developed a deep neural network for semantic segmentation, achieving consistent results in generating accurate centerlines across different test scenarios and road conditions.
- Conducted extensive literature reviews to stay up-to-date with advances in related fields such as deep learning and computer vision for HD Map Generation.
- Participated in regular brainstorming sessions to identify potential improvements and enhancements to the existing algorithm.

SMaRC Swedish Maritime Robotics Centre — Stockholm, Sweden
Research Intern — 01/2022 - 09/2022

- Developed and implemented a novel algorithm to improve the accuracy of sonar-based pose estimation for autonomous underwater vehicles, reducing drift and achieving accuracy up to 1m.
- Performed detailed analyses on the impact of various parameters such as sensor noise, and navigation drift providing valuable insights into optimizing system performance.
- Conducted extensive experiments and simulations on multiple underwater environments to validate the effectiveness of the proposed approach.

SurgiQ Ltd — Genoa, Italy
Research Intern — 03/2020 - 07/2020

- Developed and implemented an intelligent scheduling system for a hospital, resulting in 97% efficiency.
- Conducted thorough research on inductive AI techniques to inform the development of the scheduling system, leading to its publication on AI*IA and CLIC.

Skills

Programming	Python, Git
Deep Learning	TensorFlow, JAX, GCP
Data Analysis	Pandas, MongoDB, SQL
Languages	English, Italian, Dutch (basic)

Awards

2022	**Selected at Nova**, a top 3% merit-based talent network for career development	Sweden
2020	**Hattrick Award in Artificial Intelligence**, awarded to 20% of the students [Link Award]	Sweden
2020	**Selected at Lead the Future**, mentorship with a 20% acceptance rate nationwide	Italy
2020	**Scholarship at University of Genoa 2018, 2019,2020**, ranked 4th out of 150+ students	Italy

Publications

Cademartori, L., Galatà, G., Monaco, C.L., Maratea, M., Mochi, M. and Schouten, M., **2022. An ASP-based Approach to Master Surgical Scheduling.** CILC 2022: 37th Italian Conference on Computational Logic

Alviano, M., Bertolucci, R., Cardellini, M., Dodaro, C., Galatà, G., Khan, M., Maratea, M., Mochi, M., Morozan, V., Porro, I. and Schouten, M., **2020. Answer Set Programming in Healthcare: Extended Overview.** IPS-RCRA@ AI*IA.

Projects

Real Time Object Detection | KTH — Stockholm, Sweden
Python — 05/2022 - 06/2022

- Trained and optimized deep neural networks on a dataset of 10,000 images using Python and YOLOv5s, achieving a real-time object detection speed of 45 frames per second (FPS) and a mean average precision (mAP) of 61%.
  [github.com]

Imitation learning for human-robot skill transfer | TU Delft — Delft, Netherlands
Python — 09/2021 - 12/2021

- Engineered Machine Learning-based software that accurately transferred hand movements captured by kinect to a simulated manipulator, raising accuracy rate to 91%.

Training time estimation for CNN in Cloud Distributed Systems | TU Delft — Delft, Netherlands
Python — 08/2021 - 11/2021

- Utilized predictive modeling to optimize CNN training jobs, completing 33% more within pre-defined timeline utilizing 4 nodes and 16 GB of RAM.
  [github.com]

Starcraft AI Bot Competition | KTH — Stockholm, Sweden
C++ — 04/2021 - 06/2021

- Developed behavior tree algorithm in C++ for controlling multi-agent Starcraft AI, resulting in 50% win rate with 50+ simulation matches.
  [github.com]